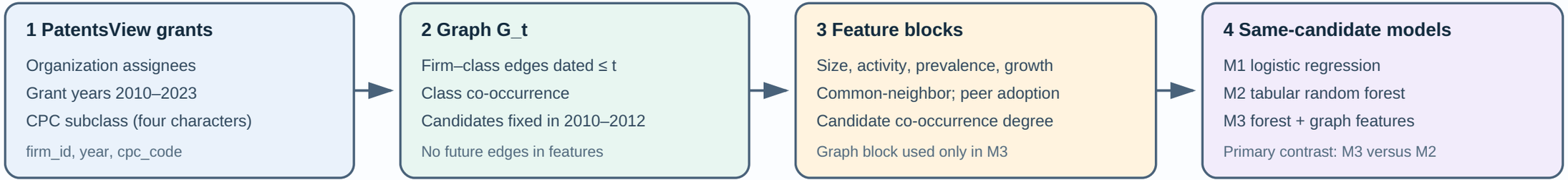
1 PatentsView grants
Organization assignees
Grant years 2010–2023
CPC subclass (four characters)
firm_id, year, cpc_code
2 Graph G_t
Firm–class edges dated ≤ t
Class co-occurrence
Candidates fixed in 2010–2012
No future edges in features
3 Feature blocks
Size, activity, prevalence, growth
Common-neighbor; peer adoption
Candidate co-occurrence degree
Graph block used only in M3
4 Same-candidate models
M1 logistic regression
M2 tabular random forest
M3 forest + graph features
Primary contrast: M3 versus M2
5 Executed holdout
Train 2013–2019 (1,540,698 pairs) · embargo 2020 · test 2021–2022 (455,458 pairs; 10,411 firms; 1.11% positives)
Primary metric: average precision. Uncertainty: firm-cluster bootstrap for M3 − M2.
Every feature at year t uses records dated ≤ t. The label is first CPC-subclass entry in t+1.